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28 October 1977
SUPERSEDING
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1 November 1973
(See 6.7)

MILITARY SPECIFICATION

ANTENNA COUPLER CU-937( )/UR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Antenna Coupler CU-937( )/UR, a remotely tuned antenna coupler intended for general purpose communications and which may be used for surface ship and fixed shore installations. The equipment operates in the 2.0 to 30.0 megahertz (MHz) frequency range and provides the means to match conventional 4.572, 7.650, or 10.698 meters (m) (15, 25, or 35 foot) whip antennas to the 100 watt peak envelope power (PEP) output of Radio Set AN/URC-35( ) or Radio Transmitter AN/URT-24( ), at the selected operating frequency.

1.2 Operational requirements. Each unit shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.i. (High-impact), Shipboard Machinery, Equipment and Systems, Requirements for (NAVY)
MIL-C-915	Cable and Cord Electrical, For Shipboard Use, General Specification For
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5985

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MIL-W-16878	Wire, Electrical, Insulated High Temperature (NAVY)
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-M-38510	Microcircuits, General Specification For
MIL-P-52192	Primer Coating, Epoxy

STANDARDS

FEDERAL

FED-STD-151	Metals; Test Methods

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirements for Enclosures For Electrical and Electronic Equipment
MIL-STD-167-1	Mechanical Vibration of Shipboard Equipment (Type 1-Environmental and Type 2 - Internally Excited)
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program

DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND

RE-B2696065	Installation Control Drawings, Data List, for AN/URC-35( )
RE-B2696865	Installation Control Drawings, Data List, for AN/URT-24( )

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. Equipments shall be in accordance with the provisions of MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3). The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design and performance. It shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the procuring activity) parts, and be completely representative of final equipment.

3.1.2 Parts, materials and processes. Parts, materials and processes shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be given to the acceptable and unacceptable materials of MIL-E-16400.

3.1.2.1 Selection of parts, materials, and processes. Selection of parts, materials, and processes shall be in accordance with MIL-E-16400. Standard parts are as indicated in MIL-E-16400.

3.1.2.2 Approval of nonstandard parts. Requests for approval for the use of nonstandard parts shall be submitted in strict accordance with the requirements of MIL-STD-965.

3.1.2.2.1 Prescreening of approval requests. Prescreening of nonstandard parts approval requests shall be accomplished in accordance with MIL-STD-965.

3.1.2.2.2 All nonstandard parts shall have been documented in accordance with the requirements of MIL-STD-965, and approved, prior to conducting the first article tests required by 4.3.

3.1.2.2.3 All approved nonstandard parts shall be subjected to quality conformance inspection, in accordance with the pertinent component specification, as specified in MIL-E-16400. Inspection shall be as specified by the procuring activity upon submission, otherwise Group A only in accordance with the pertinent component specification.

3.1.2.2.4 If any Government furnished equipment (GFE) is provided, it may include non-standard parts. The equipments furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard part approvals previously granted under earlier contracts for equipments represented by the GFE shall not be construed as being extended to equipments furnished in accordance with this specification. Non-standard parts cited in Government furnished drawings shall not be construed as being approved for use in equipments furnished in accordance with this specification.

3.1.2.3 Individual selection. The performance of the equipment shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity. If such individual part selection is approved, sufficient information and data shall be included as is necessary in all applicable technical publications and engineering drawings to enable designated repair activities to perform such selection.

3.1.2.4 Microelectronic devices. Selection and application of microelectronic devices shall comply with Requirement 64 of MIL-STD-454, requiring MIL-M-38510 Class B devices. When deviations to these requirements are necessitated due to availability or cost considerations, hermetically sealed devices inspected and tested to the requirements of MIL-STD-883, Class B may be used.

3.1.2.5 Semiconductor devices. All semiconductor devices shall be made of silicon. This was not a requirement for early models of this equipment, and may require redesign.

3.1.3 Production equipment. Production equipments shall be identical to the equipment that has been subjected to and passed the first article examinations and tests (see 3.1.1 and 4.3).

3.2 Government furnished material (GFM). Government furnished material, if any, shall be specified (see 6.2).

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3.3 Interchangeability. Interchangeability requirements, if any, shall be as specified in the contract or order (see 6.2).

3.4 Composition. The Antenna Coupler CU-937( )/UR shall consist of one each of the following:

a. Antenna Coupler unit CU-937( )/UR
b. Set of mating connectors for all receptacles

3.5 Frequency range. The frequency tuning range of the equipment shall be from 2.0 to 30.0 MHz.

3.5.1 Method of tuning. The CU-937( )/UR is tuned entirely from the associated AN/URC-35( ) or AN/URT-24( ). A tuning code is generated by the associated RF amplifier, AM-3007( )/URT, for coarse tuning, and fine tuning is accomplished by controls on the RF amplifier front panel.

3.6 Primary power source. The primary power input voltage to the equipment shall be 22 to 30 volts (nominal 28V) direct current (DC), and all primary power for control functions shall derive from the AN/URC-35( ) or AN/URT-24( ).

3.6.1 Performance limits. The equipment shall conform to the performance limits of operation under the conditions of transient voltage and power interruption as specified in MIL-E-16400 for shipboard equipment.

3.6.2 Electrical overload protection. Electrical overload protection shall be in accordance with MIL-E-16400.

3.7 Operational requirements.

3.7.1 Types of emission. Each equipment shall permit the transmission and reception of the following types of emission:

a. A1 Telegraphy without the use of a modulating audio frequency (CW)
b. A3a Telephony, single sideband (SSB), reduced carrier
c. A3b Telephony, two independent sidebands (ISB)
d. A3h Telephony, single sideband (SSB), full carrier
e. A3j Telephony, single sideband (SSB), suppressed carrier
f. A7j Single or multichannel voice frequency telegraphy (RATT)
g. A9b Two independent sidebands, combination of telephony and telegraphy (ISB/RATT). (ISB/RATT)

3.7.2 Modes of operation. The equipment shall be capable of operating with the emissions specified in 3.7.1 by selection at the associated transmitter of one of the following modes of operation:

a. LSB - Lower sideband
b. USB - Upper sideband
c. ISB - Independent sideband
d. AM - Amplitude modulation (compatible)
e. CW - Continuous wave telegraphy
f. RATT - Single or multichannel telegraphy

3.8 Equipment arrangement. The CU-937( )/UR shall be installed at the base of the associated antenna, which will normally be located 12.19m to 19.81m (40 to 65 feet) above sea level. The separation of the antenna coupler unit and the RF amplifier shall normally not exceed 91m (300 feet).

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3.8.1 The antenna coupler shall be installed on a flat mounting area having a minimum width of 30.5 centimeters (cm) (12 inches) and a minimum length of 56.0 cm (22 inches). The antenna coupler shall include suitable means, including mounting hardware, for mounting by installation personnel.

3.9 Radio transmitter system. The CU-937( )/UR is intended to be used as the antenna coupler for the AN/URC-35( ) or AN/URT-24( ). The contractor for the CU-937( )/UR shall demonstrate that the antenna coupler is operationally compatible with the AN/URC-35( ) and AN/URT-24( ), subject to availability of equipments. Mating connectors for all external cabling which require manufacture by the installing activity shall be furnished with each equipment. Additional information pertaining to installation of the equipment may be found in the Installation Control Drawings, RE-B2696065 and RE-B2696865.

3.10 Dimensions and weight. The maximum dimensions and weight of the equipment shall not exceed those shown in TABLE I:

TABLE I. Dimensions and weight.

Unit	Width Inches (cm)	Depth 1/ Inches (cm)	Height Inches (cm)	Weight Pounds (kg)
CU-937( )/UR	12 (30.5)	22 (56.0)	12 (30.5)	30 (13.6)

1/ Exclusive of connectors and mounting accessories.

3.11 Environmental service conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified herein, under those environmental conditions of MIL-E-16400 listed herein. The equipment shall operate within tolerances throughout the corresponding test specified in Section 4 without alignment or adjustment, other than external controls normally employed for operation of the equipment.

a. Operating temperature range (Range 2)
b. Non-operating temperature range
c. Humidity (0 to 100 percent)
d. Salt fog (spray) (unsheltered)
e. Shock
f. Vibration
g. Inclination (60 degrees)

3.12 Orientation. The equipment shall be capable of storage in any orientation.

3.13 Reliability.

3.13.1 Quantitative reliability requirement. The specified mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 16,000 hours for this equipment.

3.13.2 Environmental burn-in. All equipments shall be subjected to an environmental burn-in for a cumulative operating time of at least 24 hours. The equipment shall be operated as part of a AN/URT-24( ) or AN/URC-35( ) at any frequency within the specified frequency range. No adjustments of the equipment, other than normal operating adjustments, shall be allowed.

3.14 Basic conditions. The following basic conditions shall apply as pertinent to the performance requirements specified herein, except as modified by a particular test or test procedure contained in this specification. These conditions are specifically applicable to the measurement of power output and efficiency in the equipment.

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3.14.1 Power input. RF power input to the antenna coupler shall be supplied to the RF power input connector via a 50 ohm coaxial cable. All signals denoted in the following conditions shall be equivalent to those output signals supplied by the AN/URC-35( ) or AN/URT-24( ). For the purpose of comparison, all power levels denoted in the following conditions shall be equivalent to those dissipated in a standard 50 ohm dummy load under test conditions.

3.14.1.1 Single sideband suppressed carrier mode (A3j). Power input to the antenna coupler in the single sideband (SSB) mode shall be the peak-envelope-power (PEP) dissipated in a standard 50 ohm dummy load. The nominal power input to the antenna coupler shall be 100 watts PEP from 2.0 to 30 MHz.

3.14.1.2 Compatible AM mode (A3h). Power input to the antenna coupler in the compatible AM (single sideband with carrier) mode shall be the average carrier power dissipated in a standard 50 ohm dummy load. The nominal power input to the antenna coupler shall be 50 watts average from 2.0 to 30 MHz.

3.14.1.3 CW and RATT modes (A1 and A7j). Power input to the antenna coupler in the CW and RATT modes shall be the average power dissipated in a standard 50 ohm dummy load. The nominal power input to the antenna coupler shall be 50 watts average from 2.0 to 30 MHz.

3.14.1.4 Independent sideband (ISB) mode (A3b). Power input to the antenna coupler in the ISB mode shall be the same as that dissipated in a 50 ohm standard dummy load by a signal in the A3j mode, when either sideband is used alone. When both sidebands are used with carrier suppression, the total power input to the antenna coupler shall nominally not exceed that present when only one sideband is employed.

3.14.2 Voltage standing wave ratio (VSWR). The tuned VSWR, when measured at the AN/URC-35( ) or AN/URT-24( ) output on the coaxial line connected to the RF power input connector of the antenna coupler, shall not exceed 1.5:1 at any frequency from 2.0 to 30 MHz. The tuned VSWR shall be measured at any specific frequency within the frequency range of 2.0 to 30 MHz upon completion of automatic rough tuning specified in 3.16.3.1 and final manual tuning specified in 3.16.3.2.

3.14.3 Standard 50 ohm dummy load. For purpose of tests, the standard 50 ohm dummy load used during that procedure shall consist of a nonreactive 50 ohm resistance element. The dummy load shall have sufficient power handling capability so that it will dissipate 100 watts average RF power over the frequency range from 2.0 to 30 MHz under any specified operating environmental conditions, without alteration of the electrical characteristics of the resistive element.

3.15 Mechanical design and construction. Mechanical design and construction furnished in accordance with this specification shall conform to all applicable requirements of MIL-E-16400, to the extent of being compatible with the Government furnished equipment (GFE).

3.15.1 General Requirements.

3.15.1.1 Construction. Construction shall meet the requirements of MIL-E-16400. The equipment shall consist basically of a cylindrical chassis mounted on a baseplate and housed in a hermetically sealed cylindrical case.

3.15.1.2 Case. The case of the CU-937( )/UR shall be an aluminum pressurized heat-exchange type. Fastener holes on equipment face plates shall have threaded inserts. The case shall include mounting brackets to permit mounting the equipment in the proximity of the associated antenna. The case construction shall be the same as that of the GFE, except as modified by this specification.

3.15.1.3 Enclosure. The enclosure for the equipment shall be in accordance with MIL-E-16400. The degree of enclosure for the equipment shall be watertight and airtight. Pressurization shall be in accordance with 3.15.1.3.2 and the hermetic requirement as specified in MIL-STD-108.

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3.15.1.3.1 External connections. External connections shall be the same as in the GFE.

3.15.1.3.2 Pressurization. The equipment shall be capable of pressurization with 10 pounds per square inch gauge (psig) of dry nitrogen gas. The leak rate of the equipment shall be the minimum possible wherein repressurization shall not be required more than two times a year. Pressurization shall be possible using Pressurization Kit MK-260/U. An environment-proof gauge shall be provided on the outside of the equipment for internal pressure measurement. The equipment shall be provided with an over-pressure relief valve and a sign warning of over-pressurization danger. The input valve shall be of such size that should high pressure air be applied, the relief valve will provide larger volume exit and protect the equipment. The equipment, when shipped, shall be provided with a red warning tag for alerting the installing activity to pressurize the unit prior to operational use. All associated RF and control cable connectors shall be pressure proof, waterproof and noncorrosive types.

3.15.1.4 Bonding and grounding. The equipment shall conform to the requirements of MIL-E-16400, except as otherwise specified herein.

3.15.1.4.1 Case bonding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and the mounting surface. The area around each bonding area shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.15.1.4.2 Grounding stud. Grounding stud shall be a pressed-in captive insert made integral with the case and constructed of austenitic corrosion-resistant steel. The captive insert shall be flush with the outside surface of the case and shall contain a blind threaded hole to prevent the entrance of moisture into the case. A 25.4 millimeter (mm) (one-inch) diameter area around the threaded insert shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.15.1.5 Corrosion protection, corrosion resisting treatments and painting. Corrosion protection and corrosion-resisting treatments shall be in accordance with MIL-E-16400.

3.15.1.5.1 Painting. Painting shall be in accordance with MIL-E-16400, except that the interior surfaces of each metal equipment enclosure shall be painted a dull black to improve thermal efficiency.

3.15.1.5.2 Cover protection. The front cover (base plate) shall be protected from corrosion by the following:

a. Anodized in accordance with MIL-A-8625
b. Paint primer in accordance with MIL-P-52192
c. Two coats Standard Navy Gray, air-dried between coats

Front cover screws shall be painted after assembly to seal out salt collection. The pressure gauge is to be painted, exclusive of glass and the drain hole, after assembly on the panel.

3.15.1.6 Fastener hardware. Fastener hardware shall conform to the requirements of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel, except in those applications requiring material capable of withstanding high tensile stresses, or in applications where high intensity RF fields require the use of brass.

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3.15.1.7 Cables. Interconnecting cables between the antenna coupler and the RF amplifier shall be communications, nonflexing, service standard Navy cables selected from MIL-C-915. The antenna coupler shall be provided with receptacles and mating connectors for use with the selected cables. The interconnecting cables will be provided by the installing activity and are not required to be delivered with the equipment. The minimum number of conductors shall be used, and spare conductors shall be provided to the extent of compatibility with the Government furnished equipment.

3.15.1.8 Connectors. External multichannel cabling connectors shall be in accordance with MIL-E-16400. Mating plugs shall be provided with cable clamps. All external mounted connector receptacles shall be provided with a suitable protective plastic cap to prevent damage to the connector when the mating connector is not installed.

3.15.1.9 Time meter. A time meter is not required.

3.15.1.10 Motors. Motors used in the equipment shall comply with the requirements of MIL-E-16400, to the extent of compatibility with the GFE.

3.15.1.11 Airborne and structureborne noise. The airborne and structureborne noise requirements of MIL-E-16400 are not applicable.

3.15.1.12 Accessibility. In addition to the requirements of MIL-E-16400, the circuit mounting boards and components shall be easily accessible for maintenance,repair or replacement. This requirement shall be satisfied to the extent of being compatible with the Government furnished equipment.

3.15.1.13 Test provisions. Test points shall be so provided and located as to permit the isolation of malfunctioning parts or subassemblies within any major unit or assembly, without the unsoldering of any interconnection or wiring lead. Test points provided shall be identical with those extant in the GFE.

3.15.1.14 Circuit alignment. Resonant circuits shall be provided with adjustable trimmers, or equally effective means, to permit accurate alignment. The trimmers shall provide sufficient means of adjustments to compensate for the effects of normal aging of parts. If shrinkable tubing is used on variable components, it shall permit necessary adjustments to be made without having to replace the tubing.

3.15.1.15 Factory adjustment controls. The design of the equipment shall not include factory or sealed adjustment controls, unless specifically approved by the procuring activity. Where these controls are approved, a complete description and instructions for adjustment of such controls shall be included in the associated technical publications.

3.15.1.16 Shock, vibration and inclination. The equipment shall be capable of meeting the requirements specified herein without the use of resilient shock and vibration mounts.

3.15.1.16.1 Shock. The equipment shall be capable of withstanding the shock test for Grade A, deck mounted, Class I, lightweight, Type A equipment as specified in MIL-S-901.

3.15.1.16.2 Vibration. The equipment shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. The nameplate requirements of MIL-STD-167-1 are not applicable.

3.15.1.16.3 Inclination. The equipment shall conform to the inclination requirements of 3.11 and MIL-E-16400.

3.15.1.17 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the requirements of MIL-E-16400, to the extent of being compatible with the GFE.

3.15.1.18 Soldering. Soldering shall conform to the requirements of MIL-E-16400.

3.15.1.19 Welding. Welding shall conform to the requirements of MIL-E-16400.

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3.15.1.20 Attachment of wires and leads. Attachment of wires and leads shall conform to the requirements of MIL-E-16400.

3.15.1.21 Internal wiring and cabling. Internal wiring and cabling shall conform to the requirements of MIL-E-16400. All wiring shall be TFE insulated. Hook-up wire shall conform to MIL-W-16878, Type E.

3.15.1.22 Conductors. Conductors shall be in accordance with MIL-E-16400, except that the number and purpose of conductors shall be the same as those used in the GFE.

3.15.1.23 Conductor identification. Identification of conductors shall conform to the requirements of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.15.1.24 Identification and marking. Identification and marking of all units, assemblies, electrical and moving mechanical parts, and any mechanical parts required to be removed for normal maintenance, shall be in accordance with MIL-E-16400. Hertz or Hz shall be used for all indications of frequency on hardware and GFE to the maximum extent practicable. Reference designators shall be the same as used in the GFE to the maximum extent practicable.

3.15.1.25 Part replacement. Part replacement shall conform to the requirements of MIL-E-16400.

3.15.1.26 Gears and cams. Gears and cams shall conform to the requirements of MIL-E-16400, except that only metal gears shall be used.

3.15.1.27 Test cable. The equipment's technical manual shall provide instructions for manufacture of a test cable assembly that will permit maintenance testing of the antenna coupler unit in the proximity of the associated transmitter. The technical manual shall additionally provide details for a suitable antenna coupler RF output dummy load for use during this maintenance/repair period.

3.15.1.28 Switches. The design of the antenna coupler shall assure positive and long life sliding contacts and electrical limits switches to assure long-term, maintenance-free use and to prevent damage by any condition of mistuning.

3.15.1.29 Cooling. The antenna coupler unit shall provide specified operation, under continuous locked key duty, without the requirements for special external cooling facilities and without overheating or part failure.

3.15.1.30 Salt spray. The complete antenna coupler unit, including cables and mating connectors, shall comply with the requirements of MIL-E-16400 for salt spray.

3.15.1.31 Thermal design. Thermal design shall be in accordance with MIL-E-16400 to the extent of compatibility with the GFE. A thermal survey shall be performed.

3.16 Detailed performance.

3.16.1 RF power capability. The equipment shall be capable of continuous duty in a tuned condition with a nominal power level of 100 watts, over the 2.0 to 30.0 MHz frequency range. The equipment shall safely handle the maximum peak and average power outputs of Radio Set AN/URC-35( ) and Radio Transmitter AN/URT-24( ). Safety provisions shall be included in the CU-937( )/UR to preclude equipment damage while in any tuning or tuned condition and while the associated transmitter is in any mode of operation specified in 3.7.

3.16.1.1 Duty cycle. The antenna coupler shall be capable of continuous operation at full power from 2.0 to 30 MHz within the environmental extremes specified herein upon completion of rough and final tuning procedures.

3.16.1.2 Efficiency. The efficiency of the antenna coupler shall not be less than that of the GFE.

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3.16.2 Tuning method. The antenna coupler shall contain a sufficient number of variable and fixed circuit elements capable of being arranged in various circuit configurations such that the antenna coupler will afford a suitable impedance match between a 50 ohm transmission line, and any of the antennas specified in 3.16.4, from 2.0 to 30 MHz. The selection of the fixed circuit elements, and variations in the variable circuit elements, shall be accomplished by remote control signals. The tuning procedure involved in resonating a specified antenna at a particular frequency shall be divided into two operations. The first operation shall be denoted as rough or coarse tuning. The second operation shall be denoted as fine or vernier tuning. An AM signal (A3h) or carrier (A1) shall be supplied to the antenna coupler RF power input during the fine tuning operation.

3.16.3 Programming. It shall be possible to preset the coupler network configuration for a particular band of frequencies for each of the antennas specified in 3.16.4. This operation shall be denoted as programming, and initially shall be accomplished manually within the antenna coupler enclosure for each of the antenna impedances specified in 3.16.4. Programming shall be modified for each of the antenna impedances specified in 3.16.4.

3.16.3.1 Rough tuning. Rough tuning shall be initiated whenever a sufficient change is made in the coded frequency information supplied to the antenna coupler. The coded frequency information shall consist of a control signal at ground potential applied to one of the control connectors specified in 3.16.11. Upon receipt of the proper control signal, the fixed and variable elements within the antenna coupler shall position themselves in a manner so that the antenna in use shall be tuned approximately to resonance. All RF power shall be removed from the RF input connector during the rough tuning procedure.

3.16.3.2 Final tuning. Final tuning of the antenna coupler with the specified antenna shall be accomplished manually by means of the Tune and Load remote controls specified in 3.16.16, in conjunction with the remote indicators specified in 3.16.17, so that the final tuned VSWR shall lie within the limits specified in 3.16.5. During final tuning procedures, the power level of the A3h or A1 signal presented to the RF input connector on the antenna coupler for tuning purposes shall not exceed 25 watts average. Upon completion of final tuning, the RF input power to the antenna coupler may be increased to those levels specified in 3.14.1.

3.16.4 Antenna characteristics. Detailed antenna characteristics will vary with different ship types and installed locations of the antenna. The antenna characteristics of TABLE II shall be used as the normal antenna characteristics for the antennas that may be used with CU-937( )/UR.

3.16.5 VSWR. The equipment shall match any specified antenna impedance within its capability to a 50 ohm transmission line with a VSWR of not greater than 1.5:1.

3.16.6 Tuning time. The tuning time shall not be in excess of 1 minute. Tuning time shall be defined as an interval between the time at which a change is made in the coded tuning information supplied to the antenna coupler, and the time at which automatic preliminary tuning of the antenna coupler elements is completed at the new operating frequency.

3.16.7 RF power input. The antenna coupler shall operate up to RF power levels specified in 3.14.1 with AN/URC-35( ) or AN/URT-24( ), following final tuning of the antenna coupler at the operating frequency specified in 3.16.3.2.

3.16.8 Modes of coupler operation.

3.16.8.1 Normal operation. Normal operation of the antenna coupler shall involve the use of the antenna coupler singly with each of the prescribed antennas in 3.16.4 in conjunction with a single transmitter such as AN/URC-35( ) or AN/URT-24( ). Under normal conditions, the preliminary frequency to which the antenna coupler and its associated antenna are tuned shall be determined solely by the preliminary programming of the antenna coupler and coding information derived from the associated AN/URC-35( ) or AN/URT-24( ).

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TABLE II. Nominal antenna characteristic impedance.

Frequency (MHz)	Nominal Characteristic Impedance (R ± jX)					
	15 ft. whip		25 ft. whip		35 ft. whip	
2.0	6.2	-j1125	3.2	-j600	7.5	-j350
2.5	6.3	-j900	4.2	-j500		
3.0	6.5	-j750	6.0	-j400	12	-j210
3.5	6.6	-j625	7.0	-j350		
4.0	6.9	-j545	9.3	-j280	18	-j115
5.0	7.65	-j415	13	-j190	28	-j60
6.0	8.75	-j330	19	-j120	40	+j2
7.0	10.4	-j270	35	-j75	80	+j80
8.0	12.3	-j217	50	-j20	190	+j150
9.0	15.1	-j166	75	j0	580	+j80
10.0	19.9	-j125	150	+j40	300	-j200
11.0	24.3	-j85	210	+j25	120	-j290
12.0	31.75	-j52	250	-j5	50	-j200
13.0	42	-j15	250	-j110	35	-j130
14.0	56.5	+j37	210	-j130	30	-j80
15.0	71.5	+j75	180	-j160	30	-j40
16.0	97	+j123	160	-j180	33	-j8
17.0	129	+j174	130	-j190	45	+j15
18.0	174	+j213	85	-j190	70	+j40
19.0	250	+j244	55	-j170	95	+j60
20.0	389	+j260	30	-j170	120	+j70
21.0	506	+j80	20	-j210	140	-j50
22.0	467	j0	19	-j100	160	-j2
23.0	366	-j212	18	-j65	160	-j25
24.0	277	-j205	18	-j50	130	-j40
25.0	232	-j188	20	-j35	120	-j48
26.0	200	-j163	23	-j9	110	-j50
27.0	175	-j142	30	j0	110	-j50
28.0	46	-j122	35	+j30	105	-j50
29.0	41	-j106	60	+j60	100	-j30
30.0	39	-j85	110	+j95	100	-j20

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3.16.8.2 Bypass operation. Bypass operation involves the use of the antenna coupler with the antenna connected to the associated receiver antenna input in the receive condition and with the antenna coupler tuning circuits in use in the transmit condition. In accordance with 3.16.16.4 it shall be possible to bypass the tuning elements of the antenna coupler so that the antenna in use shall become directly connected to the antenna input connector on the associated receiver.

3.16.9 Overload protection. Operation of the antenna coupler and its associated antenna in close proximity to disassociated RF systems, with antennas having effective radiated power levels up to 500 watts, shall not cause damage to the antenna coupler and its components.

3.16.10 Arcing. There shall be no arcing or corona in the antenna coupler unit under any conditions of operational use throughout the frequency range of 2.0 to 30.0 MHz or with any antennas as specified in 3.16.4. Polytetrafluoroethylene insulation shall be used as required. Polyvinyl chloride insulation shall not be used.

3.16.11 Connectors.

3.16.11.1 Power and control signal connector. A single power and control signal connector shall be provided at one end of the antenna coupler. The connector shall be located on the end of the antenna coupler adjacent to the RF power input connector specified in 3.16.11.2.

3.16.11.2 RF input connector. A single RF power input connector shall be provided at one end of the antenna coupler. The connector shall be a UG-680 assembled with silicone, and shall be located on the end of the antenna coupler opposite to that on which the antenna connector is mounted. Solder connection shall be sleeved with polytetrafluoroethylene. The braid of the coaxial cable emanating from the connector shall be anchored on the adjacent front panel stud with a crimp end solder lug.

3.16.11.3 Antenna connector. A single antenna connector shall be provided at one end of the antenna coupler. The connector shall be capable of withstanding the high RF voltages present during tuning procedures. The connector shall be equipped with a suitably threaded stud to accommodate the antenna connecting lead specified in 3.16.12. The connector shall be located on the end of the antenna coupler opposite to that on which the RF input connector is mounted.

3.16.12 Separation from antenna. The antenna coupler shall be capable of being mounted in the operating environment so that the separation between the RF electrical connection at the base of the antenna in use and the terminal of the antenna connector on the antenna coupler shall be nominally 30.5 cm (12 inches). This connection shall be made with a suitable stranded copper conductor.

3.16.13 Separation from associated transmitter. The antenna coupler shall be capable of being mounted in the operating environment so that the maximum length of coaxial line and system interconnection cable between the output connectors of the associated transmitter and the input connectors on the antenna coupler shall not exceed 91.44 m (300 feet).

3.16.14 Power and control signals. The following power and control signals shall be supplied to the antenna coupler via the power and control signal connector specified in 3.16.11.1.

3.16.14.1 Rough tuning information. The rough tuning information necessary to tune the antenna coupler and antenna approximately to resonance shall be supplied by the associated transmitter in accordance with 3.16.3.1.

3.16.14.2 Power supply. The primary input voltage to the antenna coupler shall be 22 to 30 volts (nominal 28V) DC and all primary power for control functions shall derive from the Radio Set AN/URC-35( ) or Radio Transmitter AN/URT-24( ).

3.16.15 Outputs. Output signals shall be provided by the antenna coupler to the associated transmitter via the power and control signal connector specified in 3.16.11.1.

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3.16.15.1 Antenna interlock. An interlock signal shall be made available so that an indication on one of the conductors at the power and control signal connector specified in 3.16.11.1 shall be made when the proper cable is mated to the connector, provided that primary power is available.

3.16.15.2 Antenna tune. An antenna tune signal shall be made available so that an indication on one of the conductors at the power and control signal connectors specified in 3.16.11.1 shall be made during rough and fine tuning procedures.

3.16.16 Controls. Controls shall be provided for proper operation of antenna coupler in accordance with this specification as follows.

3.16.16.1 Location. The controls shall be located on the associated transmitter. Connection between the controls and the antenna coupler shall be accomplished via the power and control signal connector.

3.16.16.2 Tune. There shall be a switch labeled TUNE. Adjustment of this control shall provide fine tuning of the antenna coupler in conjunction with the load control specified in 3.16.16.3.

3.16.16.3 Load. There shall be a control labeled LOAD. Adjustment of this control shall provide fine tuning of the antenna coupler in conjunction with the TUNE control specified in 3.16.16.2.

3.16.16.4 Antenna coupler bypass. There shall be a switch labeled ANTENNA COUPLER BYPASS. Adjustment of this switch shall cause the antenna connector on the antenna coupler to become directly connected to the antenna input connector on an associated receiver so that all tuning elements within the antenna coupler shall be disconnected from the antenna connector.

3.16.16.5 Meter selector switch. There shall be a switch labeled FORWARD POWER, POWER OUTPUT, and TUNE (reflected power). This switch shall normally rest in the center POWER OUTPUT position. Operation of the switch to the FORWARD POWER and TUNE (reflected power) positions shall cause the RF POWER METER to read the designated parameters in normal operation.

3.16.17 Indicators. Indicators shall be provided for proper operation of the antenna coupler in accordance with this specification as follows.

3.16.17.1 Location. The indicators shall be provided on the associated transmitter. Connection between the indicators and the antenna coupler shall be accomplished via the power and control signal connector specified in 3.16.11.1.

3.16.17.2 Antenna tune. There shall be a pilot lamp indicator labeled ANTENNA TUNE which shall provide a visual indication whenever the rough or fine antenna tuning procedures are initiated. Control circuitry for this indicator shall be included in the antenna coupler.

3.16.17.3 Power meter. There shall be a suitable meter labeled RF POWER which shall indicate the parameters designated by the meter selector switch in 3.16.16.5.

3.16.18 Electromagnetic compatibility (EMC). The equipment shall meet the requirements of MIL-STD-461 for Class IC, C-E equipment. Tests CE01 and CE03 shall be performed.

3.16.19 Accelerated life. The equipment shall conform to the requirements of MIL-E-16400 for accelerated life.

3.17 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of product to be offered for acceptance; c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.2 General inspection. Inspection shall be as specified in MIL-E-16400 except as otherwise specified herein.

4.2.1 Classification of inspection. The methods of examination and testing of electronic equipment fall within the following classifications:

a. First article inspection (see 4.3)
b. Quality conformance inspection

    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
    3. Environmental inspection (see 4.4.3)

c. Reliability testing (see 4.10)
d. Preparation for delivery inspection (see 4.11)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with all applicable requirements of this specification, including the tests specified in TABLE III (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection (group A). Production inspection shall be made on every item offered for delivery. This inspection shall comprise such examination and tests which will prove the workmanship and reveal omissions and errors of the production process such as: functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls/adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests shown in group A of TABLE V.

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TABLE III. Examination and tests.

Examination or test	Requirement	Test	First Article	Q C INSP A	B	C
Surface examination	3.1.2	MIL-E-16400	X	X		
Weight	3.10	MIL-E-16400	X	X		
Size	3.10	MIL-E-16400	X	X		
Parts and materials	3.1.2	MIL-E-16400	X	X		
Finish	3.1.2	MIL-E-16400	X	X		
Marking	3.15.1.24	MIL-E-16400	X	X		
Workmanship	3.17	MIL-E-16400	X	X		
Weld	3.15.1.19	MIL-E-16400	X		X	
Mechanical design	3.15	MIL-E-16400	X		X	
Pressurization	3.15.1.3.2	4.9.9		X		
Bonding and grounding	3.15.1.4	MIL-E-16400	X		X	
Corrosion protection	3.15.1.5	MIL-E-16400	X		X	
Painting	3.15.1.5.1	MIL-E-16400	X		X	
Soldering	3.15.1.18	MIL-E-16400	X		X	
Conductors/identification	3.15.1.22	MIL-E-16400	X		X	
Power output	3.16.1.1	4.9.1	X	X		
Efficiency	3.16.1.2	4.9.2	X		X	
VSWR	3.16.5	4.9.3	X	X		
Tuning time	3.16.6	4.9.4	X		X	
Bypass operation	3.16.8.2	4.9.5	X		X	
Arcing	3.16.10	4.9.6	X		X	
Operational test	3.16.1	4.9.7	X	X		
Antenna separation	3.8	4.9.8	X		X	
Steady state voltage	3.6	4.9.14	X		X	
Power/power factor	3.6	4.9.16	X		X	
Steady state frequency	3.6.1	4.9.14	X		X	
Transient voltage	3.6.1	4.9.14	X		X	
Transient frequency	3.6.1	4.9.14	X		X	
Power interruption	3.6.1	4.9.15	X		X	
Temperature sequence	3.11	4.9.11	X			X
Low temperature	3.11	4.9.11	X			X
High temperature	3.11	4.9.11	X			X
Humidity	3.11	4.9.11	X			X
Salt fog (spray)	3.11	MIL-E-16400	X			X
Accelerated life	3.16.19	4.9.17 and MIL-E-16400	X			
Reliability	3.13	4.10				
Enclosure	3.15.1.3	4.9.13	X			X
Shock	3.15.1.16.1	4.9.12	X			X
Vibration	3.15.1.16.2	4.9.12	X			X
Inclination	3.15.1.16.3	4.9.12	X			X
Thermal Design	3.15.1.30	MIL-E-16400	X			
EMI	3.16.18	4.9.10	X			X
Environmental burn-in	3.13.2	4.10.3		X		

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4.4.2 Production control inspection (group B). Production control inspection shall be conducted on a sampling basis as specified. This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examination and tests shown in group B of TABLE III. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1. Sampling shall be in accordance with MIL-STD-105, using inspection level S-3 with an acceptable quality level (AQL) of 6.5 percent for each attribute.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.3 Environmental inspection (group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration and inclination. Environmental inspection shall include the examination and tests shown in group C of TABLE III.

4.4.3.1 Sampling for inspection of equipment. For the first environmental inspection, one equipment from the first month's production, or from the first 50 units produced, whichever is less, shall be selected. For subsequent environmental inspections, one equipment from each successive 100, or one per month if less than 100 are produced monthly, shall be selected. EMI testing shall be conducted only on one of each successive 200 equipments. Tests shall be conducted on lots that have passed the production inspection of 4.4.1.

4.4.3.2 Noncompliance. If a sample unit fails environmental inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, processes, and so forth. If the Government inspector considers that the corrective action will not enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.5).

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections, or both, may be accepted on the contract provided that they are resubjected to and pass production inspection after repair of all defects.

4.5 Test plan and test procedures. Test plans and test procedures shall be as specified (see 6.2.2).

4.6 Test reports. Test reports shall be as specified (see 6.2.2).

4.7 Test facilities, chambers and apparatus. Test facilities, chambers and apparatus shall be in accordance with MIL-E-16400.

4.8 Performance of tests. Performance of tests shall be in accordance with MIL-E-16400 and as specified herein.

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4.9 Test methods.

4.9.1 RF power. The equipment shall be operated with 50 watts average power continuous locked key input into dummy loads having the characteristics as specified in TABLE II at the frequencies of 2.0, 4.0, 8.0, 12.0, 24.0, and 29.0 MHz as a minimum.

4.9.2 Efficiency. Measurements shall be made to verify compliance with 3.16.1.2.

4.9.3 VSWR. Measurements shall be made to verify compliance with 3.16.5.

4.9.4 Tuning time. Measurements shall be made to verify compliance with 3.16.6.

4.9.5 Bypass operation. Tests shall be performed to verify compliance with 3.16.8.2.

4.9.6 Arcing. Tests shall be performed to verify compliance with 3.16.10.

4.9.7 Operational tests. The equipment shall successfully complete a continuous operational test with 50 watts average power input, at 2.0 MHz into a 10.698 m (35 foot) whip antenna dummy load for a continuous period of two hours without damage or degradation of the CU-937( )/UR equipment.

4.9.8 Antenna separation. Measurements shall be made to verify compliance with 3.8.

4.9.9 Pressurization. Leakage shall be tested using the Federal Test Method Standard 151, Method 442. Pressure shall not be reduced below 8 psig.

4.9.10 Electromagnetic compatibility test. Compliance with 3.16.18 shall be verified by tests performed in accordance with MIL-STD-462.

4.9.11 Temperature and humidity. Tests shall be conducted in accordance with MIL-E-16400. Measurements of power output shall be made. During the last hour of the high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled to 65°C by radiation-baffled heating and cooling sources as required. This test is intended to simulate the worst case shipboard thermal environment (see 4.9.11.1). During humidity tests, measurements shall be made at the end of the second and fifth cycles.

4.9.11.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient is visible to the unit under test. The maximum allowable temperature difference between any two points in the chamber shall be 10°C. The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant within ±1°C of any selected test temperature.

4.9.12 Shock, vibration and inclination. Tests of operational capability shall be conducted in accordance with 4.9.1.2 before and after completion of shock and inclination tests and before, during and after the vibration tests, in accordance with 3.15.1.16. Continuous key down operation is required during all tests, unless otherwise specified herein.

4.9.13 Enclosure. Tests shall be conducted in accordance with MIL-E-16400. Measurement of power output shall be made prior to and after testing. Measurements shall be within the limits established in 3.16.1.1.

4.9.14 Supply line voltage and transients. Tests shall be performed in accordance with MIL-E-16400. Reference power output measurements shall be made prior to start of testing. Variation of power output with line voltage shall be measured against the requirements of 3.6. Following performance of the tests specified in MIL-E-16400, no permanent changes shall be produced in power output.

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4.9.15 Power interruption. In addition to the tests specified in MIL-E-16400, a test shall be conducted to verify operation with high voltage transients of three and one-half times nominal voltage but of short duration. No permanent changes shall be produced in power output as a result of this test.

4.9.16 Power and power factor. Tests shall be conducted in accordance with MIL-E-16400.

4.9.17 Accelerated life. Measurements of power output shall be made in accordance with 4.9.1 in the 2, 17, and 29 MHz bands to verify compliance with MIL-E-16400.

4.10 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781. The reliability demonstration shall be performed after first article approval.

4.10.1 Reliability demonstration (qualification) tests. Reliability demonstration testing shall be performed on at least six production units. Testing shall be performed at Test Level C with Test Plan IV of MIL-STD-781. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for Test Plan IV, using a specified MTBF (θ_0) of 2000 hours. Reliability demonstration shall be successfully completed prior to delivery of any production units under the contract.

4.10.1.1 Failure. Failure shall be any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements specified herein (for definition of failure, see 6.6.1). Any adjustments or alignments, after test initiation shall be considered as a malfunction.

4.10.1.2 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer design, corrective action shall be performed on all equipment delivered under the contract.

4.10.1.3 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to ensure that it conforms to the requirements of this inspection. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it conforms to all terms and conditions of the contract for acceptance.

4.10.1.4 Reliability test procedures. The reliability test shall be conducted using the procedures, schedules, and performance limits specified herein, as a minimum.

4.10.1.4.1 Input line voltage. Input line voltage shall be as specified in 3.6. Input voltage cycling is not required.

4.10.1.4.2 Test frequencies. During the reliability tests, at least one frequency shall be tested in each band as shown in TABLE IV. Frequencies shall be alternated on a daily basis.

TABLE IV. Test frequencies.

F1	2.0015 MHz
F2	5.3340 MHz
F3	10.6655 MHz
F4	29.9990 MHz

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4.10.1.4.3 Test schedule. Antenna couplers shall be exercised during operating on periods using a voice modulated RF input signal producing 100 watts PEP. At the contractor's option, operating duty cycles specified for reliability testing by MIL-T-24038 or MIL-R-28707 or both, may be used in lieu of the mode specified herein.

4.10.1.4.4 Minimum performance limits. At least once per day, the contractor shall provide necessary tests to verify the following performance characteristics:

a. Power output
b. VSWR
c. Rough and fine tuning functions

4.10.1.5 Burn-in. The contractor may elect to operate the reliability test samples for a certain time prior to the initiation of reliability demonstration testing, in addition to the burn-in period specified in 3.13.2. Any such burn-in applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered nonrelevant but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the reliability demonstration test samples.

4.10.2 Equipment burn-in. All deliverable equipments and spares shall be subjected to the temperature burn-in requirements of 3.13.2. The burn-in shall be performed at a stabilized 50°C ambient temperature with 1.0g ±10 percent vibration applied at least 10 minutes out of each operating hour for a cumulative operating time of at least 24 hours. Failures occurring during the last 24 hours shall be repaired and burn-in resumed for a minimum of 24 hours without further failure.

4.10.2.1 Equipment operation during burn-in. Equipment operation need not be continuous as long as the accumulated operating time conforms to conditions of 4.10.2. The equipment shall be operated in accordance with 3.13.2.

4.10.2.2 Performance monitoring. Equipment shall be checked for proper performance at least twice every 24 hours and just prior to end of burn-in.

4.10.3 Failure analysis. All failures shall be analyzed and the results of analysis shall be retained by the contractor and be available for Government review as required.

4.11 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order).

5.1 Preservation, packing, packaging and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2).

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6. NOTES

6.1 Intended use. The antenna coupler, CU-937( )/UR, is intended for general purpose voice, CW and radio teletypewriter communications, and may be used for surface ship, submarine and fixed shore installation.

6.2 Ordering data. Procurement documents should specify the following.

6.2.1 Procurement requirements.

a. Title, number and date of this specification
b. Quantity of first article sample units required (see 3.1.1)
c. Government furnished material (see 3.2)
d. Interchangeability requirement, if required (see 3.3)
e. Levels of preservation, packaging, packing and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.5	Test plans	DI-T-3709
4.5	Test procedures	UDI-T-22710

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of at least one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Nomenclature. Nomenclature referred to herein and those assigned to the GFE units may be subject to change on any contract resulting in equipment produced in accordance with this specification.

6.5 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.6 Standard definitions.

6.6.1 Failure. Failure is defined as any malfunction or parameter deviation that prevents equipment from performing within the specified operational requirements.

6.7 Changes from previous issue. The extent of changes (deletions, additions, and so forth) precludes the annotation of the individual changes from the previous issue of this document.

Preparing activity:
NAVY-EC
(Project 5985-N476)

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